
포트폴리오

(랜더링)



2026. 05. 5
권영일

Rendering Engine(SDK)

Scene

Two-Color Background, Platform, Axis, Build Envelop, Selection, Rotation, Panning, Zoom, Clipping

Primitive Type

DPipe, CDPolygon2D, CDPolygons2D, CDCylinder, CDCloud, CDText, CDRectangle, CDPedestal, CDMultiLine, CDMultiLines, CDRuller, CDGridRuller, CDGridDeepenExtend, CDCoordinateSystem, CDCoordinateSystem2D, DPolyhedron, DModel, DPoint, DLine, DTriangle, DCircle, DHexahedron, DCube, DArrow2D

Application

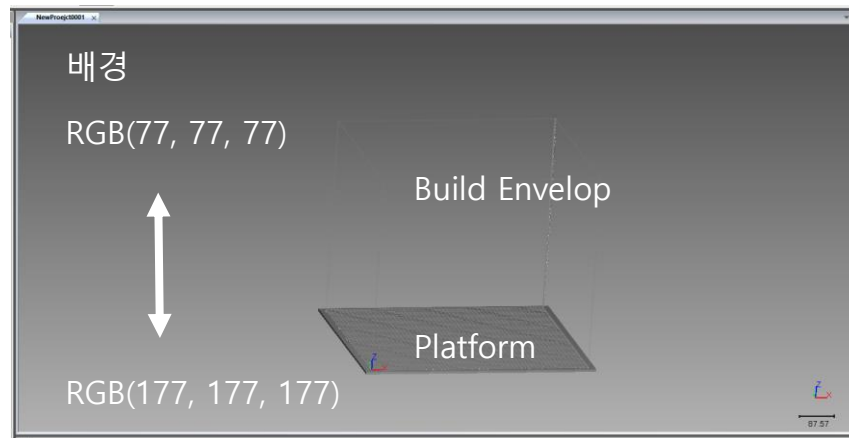
MyD, CubiCreatorLP v1~v2, HyInspector, ReshaperSolution

Rendering : Scene

<https://github.com/yikwon9801/3DReshaperSDK/blob/main/Reshaper3DRenderingEngine/Display/View/DOpenGLScene.h>

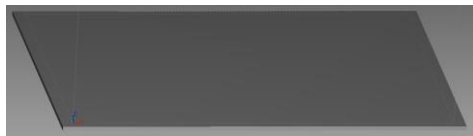
Two-Colors Background

- Current
Top Color : RGB(77, 77, 77)
Bottom Color : RGB(177, 177, 177)
- Gradient



Platform

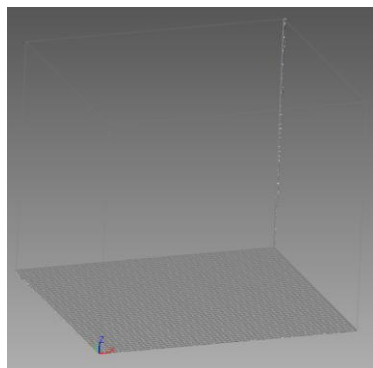
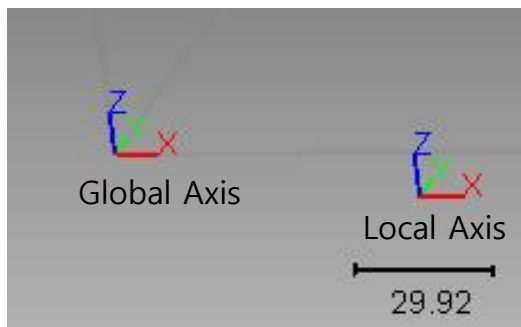
- Current
Size : 562.5x500



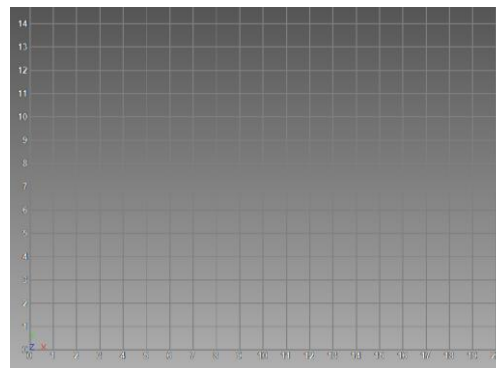
Build Envelop

- Current
Size : 562.5x500x500

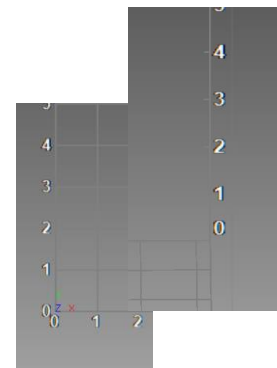
Axis



Build Bounding Box



Grid



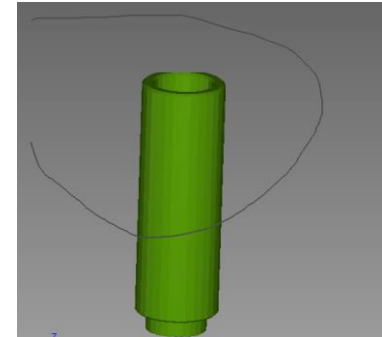
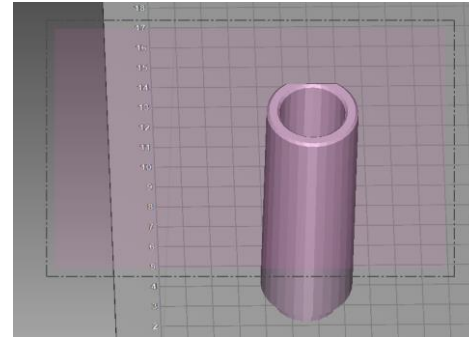
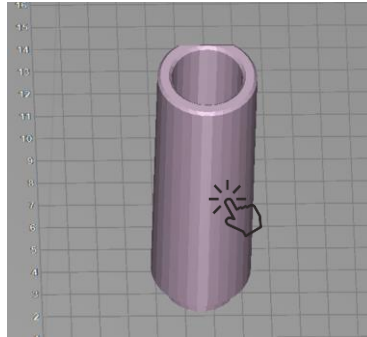
Ruler

Rendering : Scene

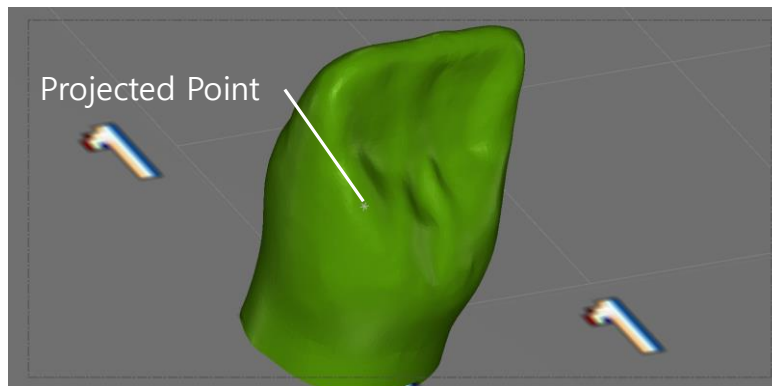
<https://github.com/yikwon9801/3DReshaperSDK/blob/main/Reshaper3DRenderingEngine/Display/View/DOpenGLScene.h>

Selection : 3가지 방식

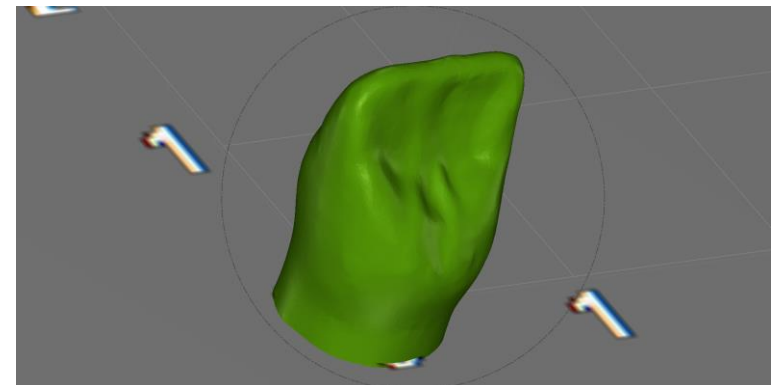
- 좌측 마우스 클릭
(사각형)
- 좌측 마우스 드래그
(사각형)
- 최측 마우스 드래그
(자유 곡선)



Rotation : 2가지 방식



- 매쉬 상 클릭된 점을 기준으로 화면 회전
(Rotation from Projected Point on Mesh)

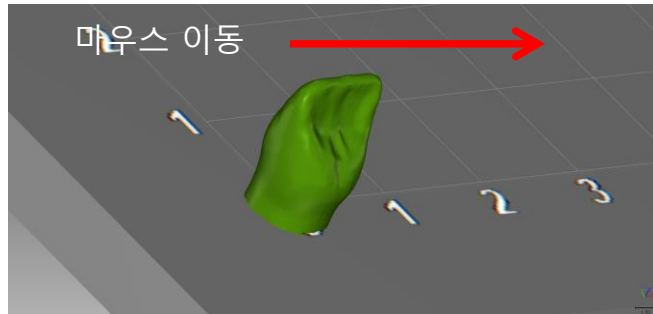


- 보이는 뷰의 평면을 기준으로 2D 회전
(2D Rotation on Screen Plane)

Rendering : Scene

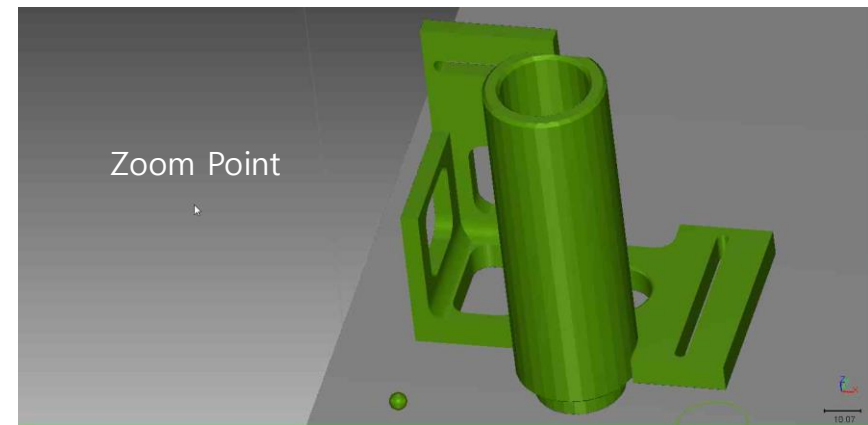
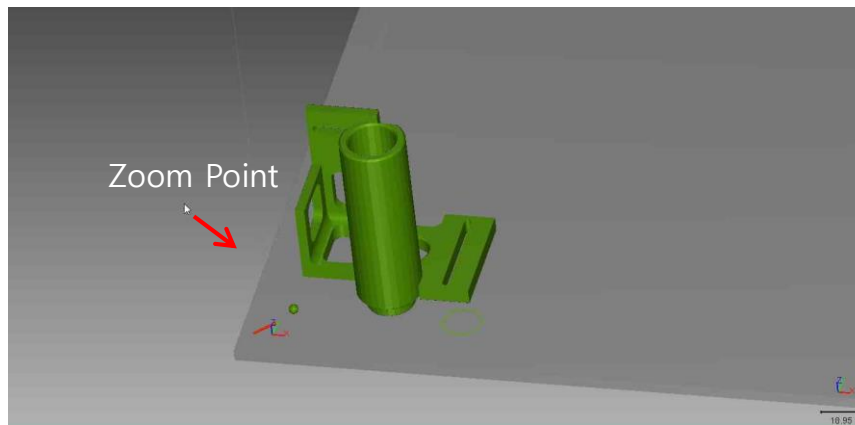
<https://github.com/yikwon9801/3DReshaperSDK/blob/main/Reshaper3DRenderingEngine/Display/View/DOpenGLScene.h>

Panning : 1가지 방식



- 중앙 휠 마우스 드래그

Zoom In/Out : 1가지 방식

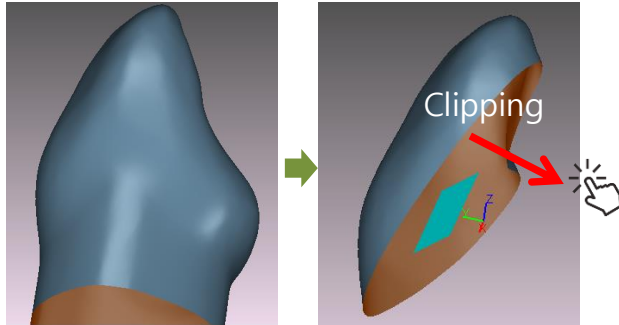


- 중앙 휠마우스 스크롤 : 현재 마우스 포인터 기준으로 확대/축소

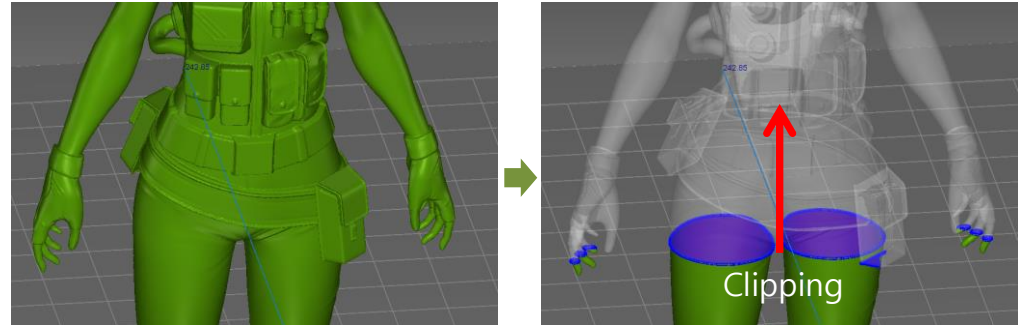
Rendering : Scene

<https://github.com/yikwon9801/3DReshaperSDK/blob/main/Reshaper3DRenderingEngine/Display/View/DOpenGLScene.h>

Clipping

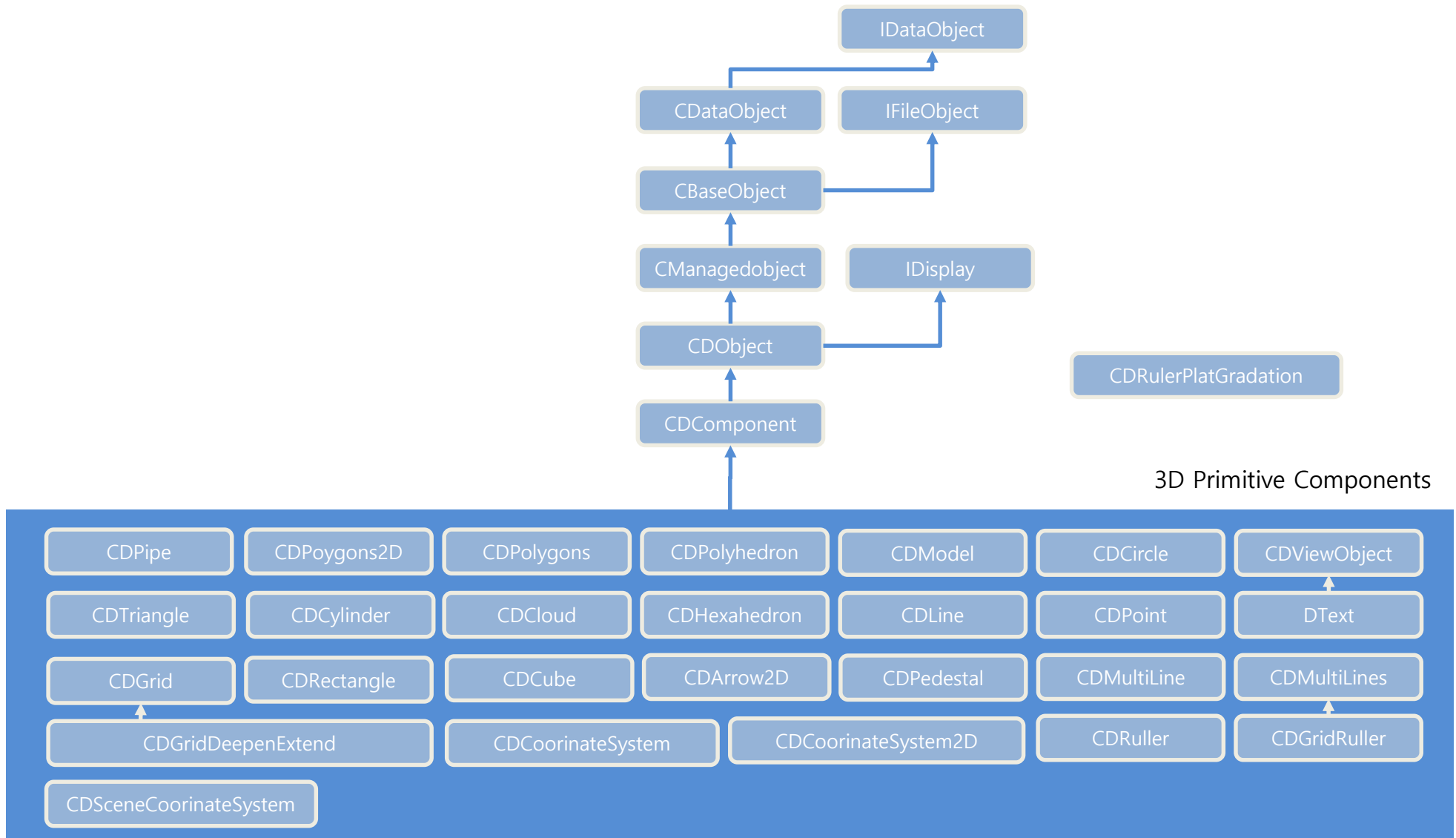


- Plane Clipping



- Z Clipping

Rendering : Primitive type : Hierachy

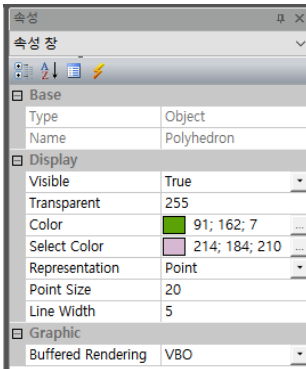


Rendering : Primitive type : DPolyhedron

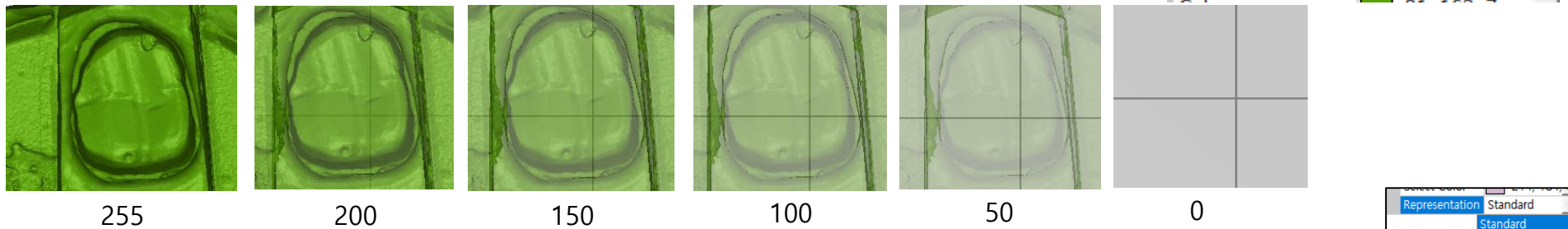
3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/Mesh/DPolyhedron.h

Attribute

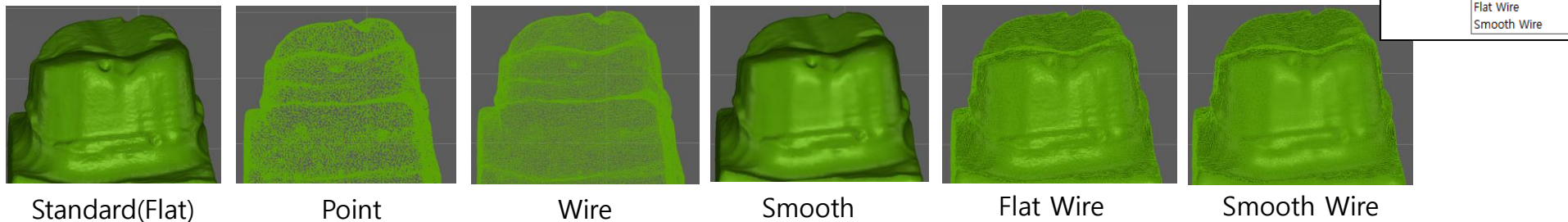
- Visible : 보기/숨김
- Transparent : 투명도
- Color : 색상
- Select Color : 선택 색상
- Representation : 표현 방법
- Point Size : Point 표현 방법으로 보기시 점의 크기
- Line Width : Wire 표현 방법으로 보기시 라인의 두께
- Buffered Rendering : GPU와의 데이터 전송 방식



Transparency(Alpha value)



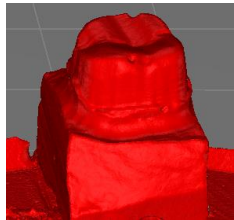
Representation



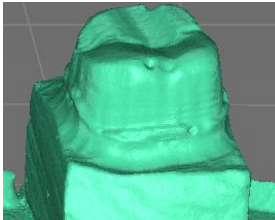
Rendering : Primitive type : DPolyhedron

[3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/Mesh/DPolyhedron.h](#)

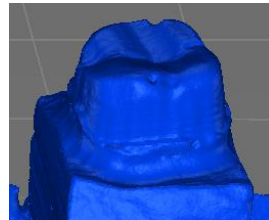
Color



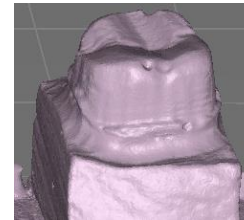
Color 255; 0; 0



Color 87; 255; 192



Color 0; 70; 255



Color 214; 184; 210

transparent	0
Color	0; 70; 255
Select Color	214; 184; 210

Face & Back

Color 91; 162; 7



GL_MATERIAL(GL_FRONT, GL_BACK)

Face : Specular {0.200000003, 0.200000003, 0.200000003, 0.00000000}, Shininess (52.1),

Emission {0.00000000, 0.00000000, 0.00000000, 0.00000000}

Back : Specular {0.00000000, 0.00000000, 0.00000000, 0.00000000}, Shininess (0),

Emission : Color Inverted

GL_MATERIAL(GL_FRONT_AND_BACK)

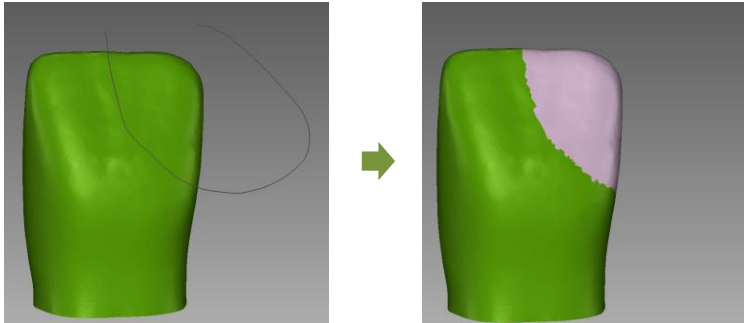
Face & back : Specular {0.200000003, 0.200000003, 0.200000003, 0.00000000}, Shininess (52.1),

Emission {0.00000000, 0.00000000, 0.00000000, 0.00000000}

Rendering : Primitive type : DPolyhedron

Partial(Selected) Color

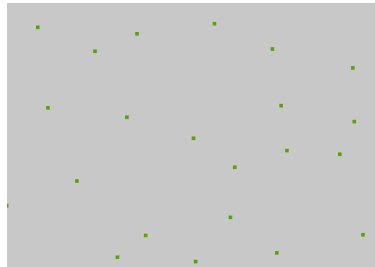
[3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/Mesh/DPolyhedron.h](#)



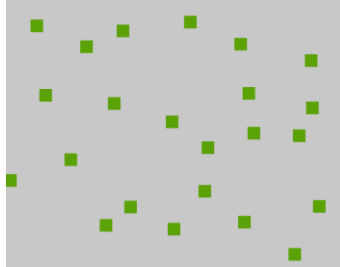
Select Partial with free-line

Selected Faces

Point Size

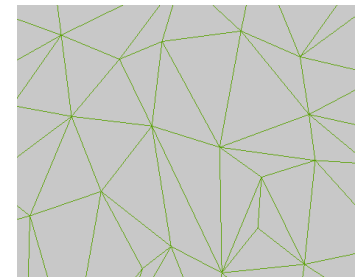


5

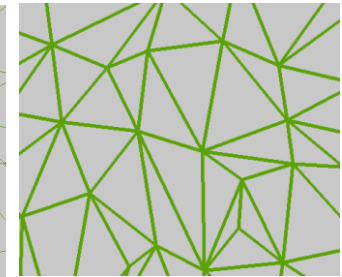


20

Line Width

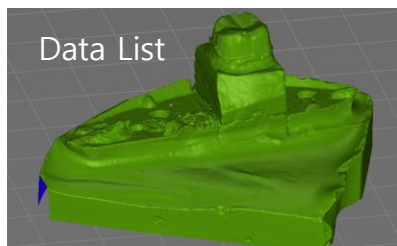


1



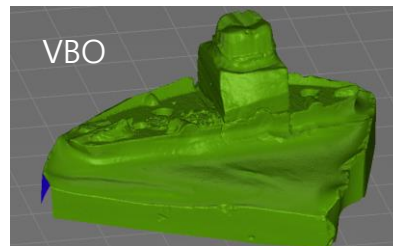
5

Graphic Buffered(Data List, VBO)



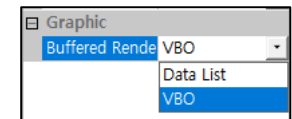
Data List

```
::glGenList
::glNewList
::glEndList
::glCallList
```



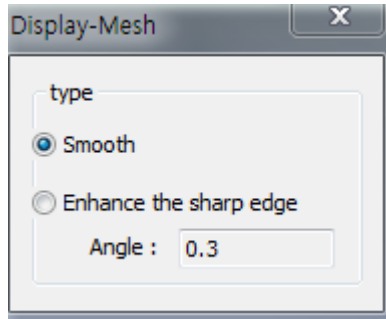
VBO

```
::glGenBuffer
::glBindBuffer
::glMapBuffer
::glDrawElements
```

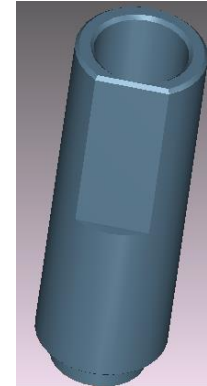


Rendering : Primitive type : DPolyhedron

Enhance the sharp edge



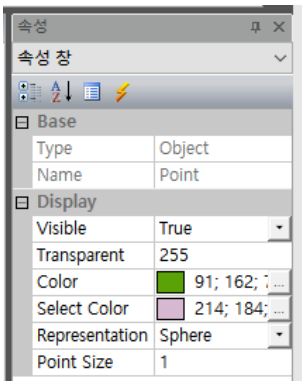
Smooth



Sharp Edge

Enhance the sharp edge 버튼을 누른다.

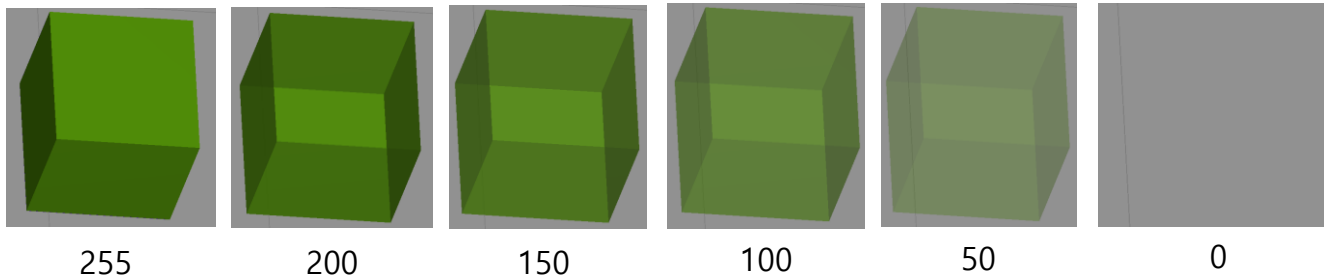
Rendering : Primitive type : DPoint

[3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/DPoint.h](#)


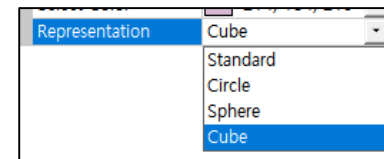
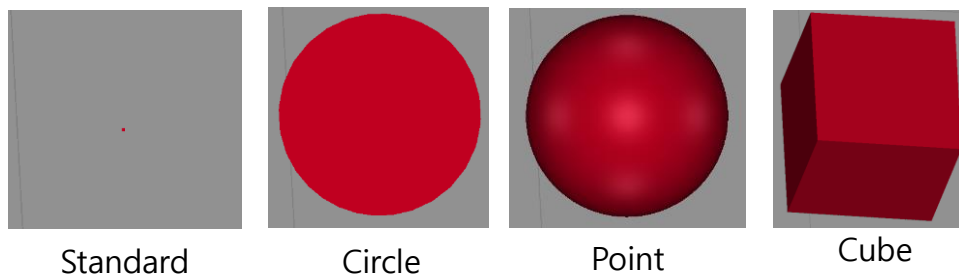
Attribute

- Visible : 보기/숨김
- Representation : 표현 방법
- Transparent : 투명도
- Point Size : 점의 크기
- Color : 색상
- Select Color : 선택 색상

Transparency(Alpha value)



Representation

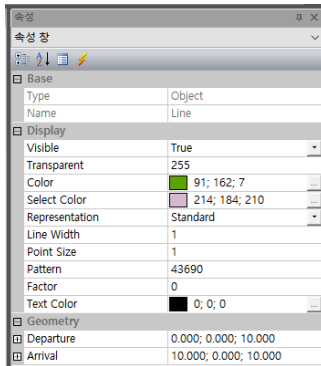


Rendering : Primitive type : DLine

[3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/DLine.h](#)

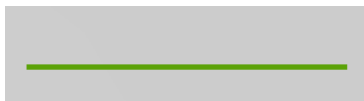
Attribute

- Visible : 보기/숨김
- Transparent : 투명도
- Color : 색상
- Select Color : 선택 색상
- Representation : 표현 방법
- Line Width : 라인 표현 방법에서 라인의 두께
- Point Size : 용도 없음
- Pattern & Factor : 점선 패턴 모양과 패턴 크기
- Text & Text Color : 텍스트 표현 방법에서 텍스트, 색상
- Departure, Arrival : 시작점과 도착점



Representation

Representation	Standard with Text
Line Width	Standard
Point Size	Standard with Text



Standard



Standard with Text

Pattern & Factor

Pattern	00ff
Factor	1



Pattern : 0x00FF



Scale : 2

Text & Text Color

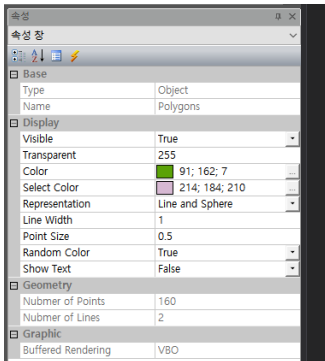
Text	Text
Text Color	255; 0; 0



Text

Rendering : Primitive type : DPolygons

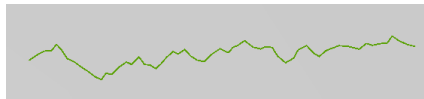
3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/DPolygons.h



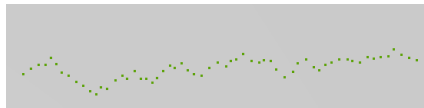
Attribute

- Visible : 보기/숨김
- Transparent : 투명도
- Color : 색상
- Select Color : 선택 색상
- Representation : 표현 방법
- Line Width : 라인 표현 방법에서 라인의 두께
- Point Size : 점 표현 방법에서 점의 크기
- Number of Points : 점의 개수
- Number of Lines : 선의 개수

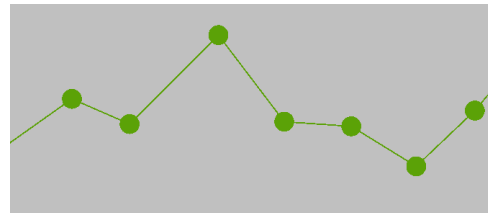
Representation



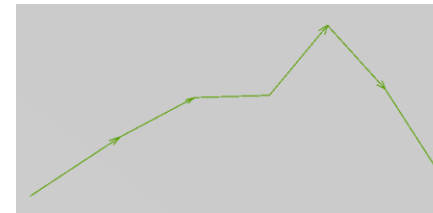
Standard



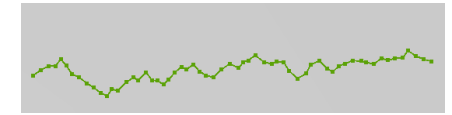
Point



Multiline with Circle



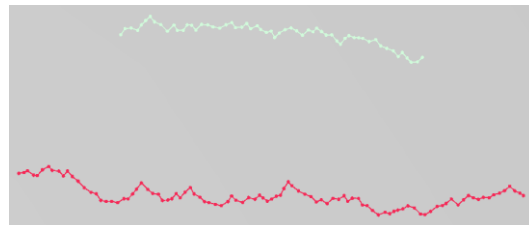
Multiline with Arrow



Multiline with Point

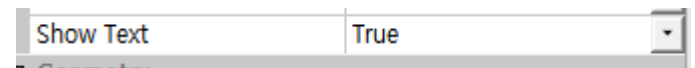
Representation	Standard
Line Width	Standard
Point Size	Point
Geometry	Multiline with Point
Number of Points	Multiline with Arrow
Number of Lines	Line and Sphere

Random Color



Random Color(Polygon)

Show Text



Text : Index, Coordinate

Rendering : Primitive type : DRectangle

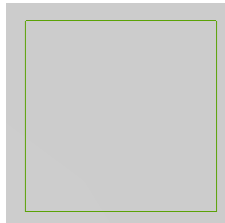
[3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/DRectangle.h](#)

속성 창	
Base	
Type	Object
Name	Rectangle
Display	
Visible	True
Transparent	255
Color	91; 162; 7
Select Color	214; 184; 210
Representation	Standard
Line Width	1
Geometry	
Point1	10.000; 10.000; 10.000
Point2	20.000; 10.000; 10.000
Point3	20.000; 20.000; 10.000
Point4	10.000; 20.000; 10.000

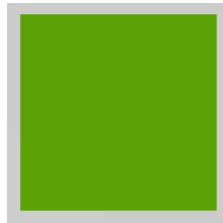
Attribute

- Visible : 보기/숨김
- Representation : 표현 방법
- Transparent : 투명도
- Line Width : 라인 표현 방법에서 라인의 두께
- Color : 색상
- Point1~4 : 사각형의 각 꼭지점
- Select Color : 선택 색상

Representation



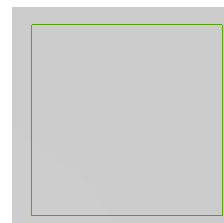
Standard



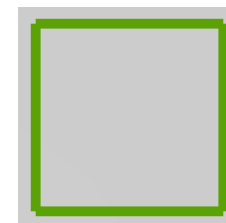
Face

Representation	Face
Line Width	Standard
Resolution	Face

Line Width



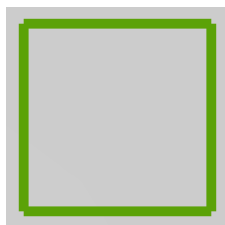
1



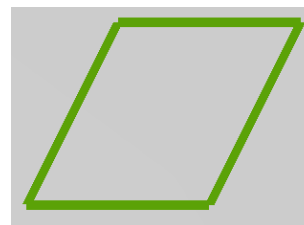
10

Representation	Face
Line Width	10
Resolution	100

Point1~4



Point1	15.000; 10.000; 10.000
Point2	25.000; 10.000; 10.000
Point3	25.000; 20.000; 10.000
Point4	15.000; 20.000; 10.000



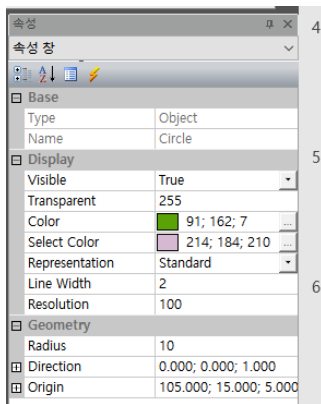
Point1	5.000; 10.000; 10.000
Point2	15.000; 10.000; 10.000
Point3	20.000; 20.000; 10.000
Point4	10.000; 20.000; 10.000

Rendering : Primitive type : DCircle

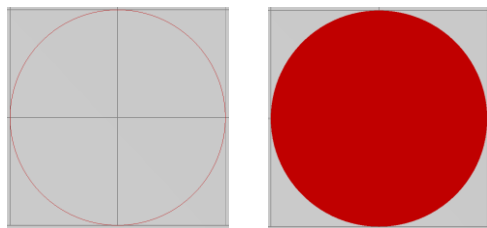
[3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/DCircle.h](#)

Attribute

- Visible : 보기/숨김
- Transparent : 투명도
- Color : 색상
- Select Color : 선택 색상
- Representation : 표현 방법
- Line Width : 라인 표현 방법에서 라인의 두께
- Resolution : 원의 선분의 개수
- Radius : 원의 반지름
- Direction : 원의 방향
- Origin : 원의 중점

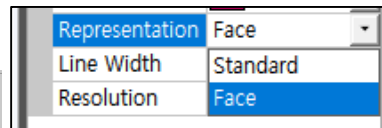


Representation

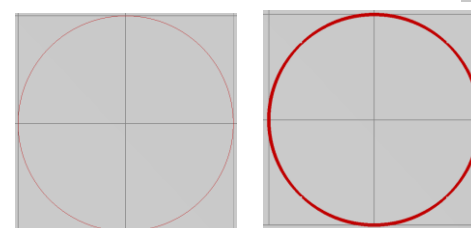


Standard

Face

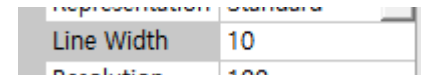


Line Width

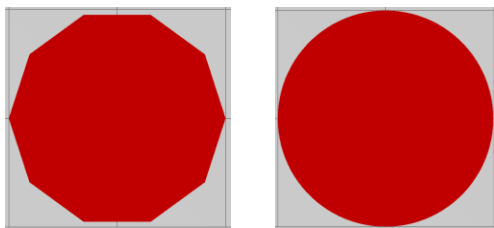


1

10

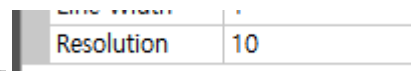


Resolution

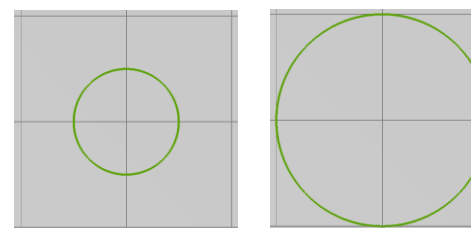


10

100



Radius

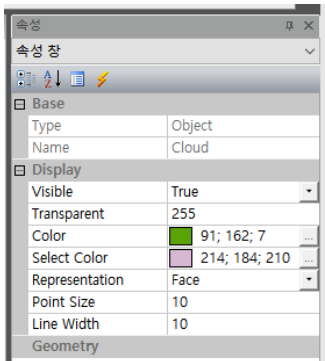


5

10



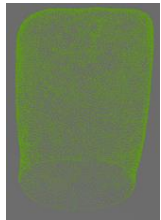
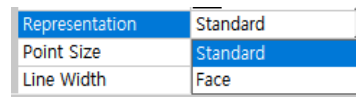
Rendering : Primitive type : DCloud

3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/DCloud.h


Attribute

- Visible : 보기/숨김
- Transparent : 투명도
- Color : 색상
- Select Color : 선택 색상
- Representation : 표현 방법
- Point Size : 스탠다드 표현 방법에서 점 크기
- Line Width : 라인 표현 방법에서 라인의 두께

Representation

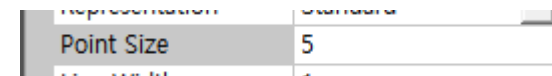


Standard

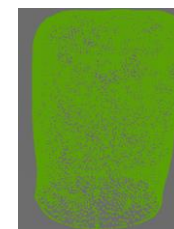


Line

Point Size

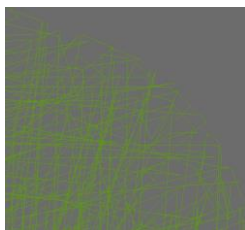


1

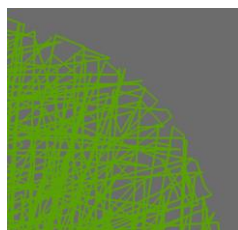


5

Line Width



1



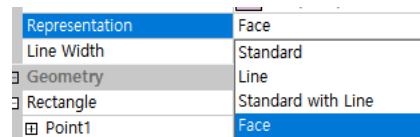
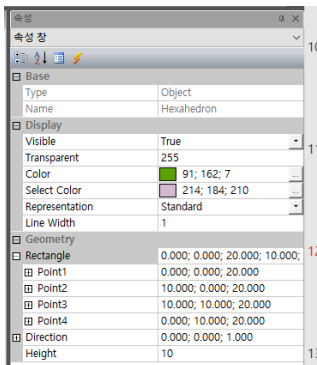
5

Rendering : Primitive type : DHexahedron

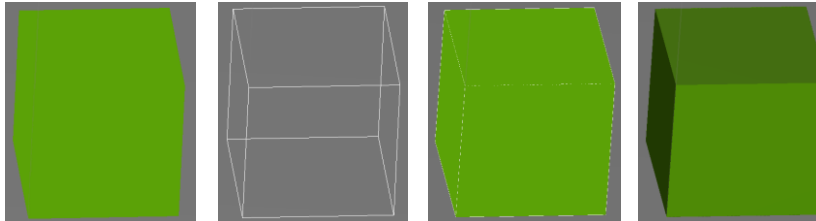
[3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/DHexahedron.h](#)

Attribute

- Visible : 보기/숨김
- Transparent : 투명도
- Color : 색상
- Select Color : 선택 색상
- Representation : 표현 방법
- Line Width : 라인 표현 방법에서 라인의 두께
- Rectangle : 육면체상에 사각형영역
- Direction : 육면체의 방향
- Height : 육면체의 높이



Representation



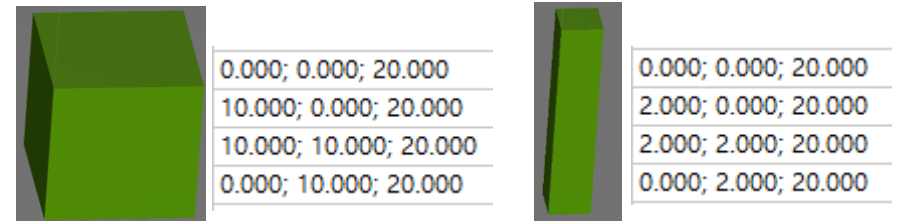
Standard

Line

Standard + Line

Face

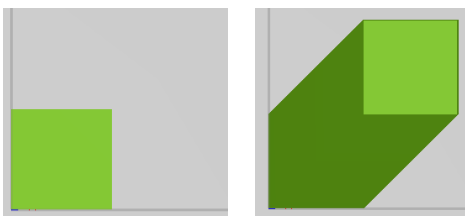
Rectangle



10x10

2x2

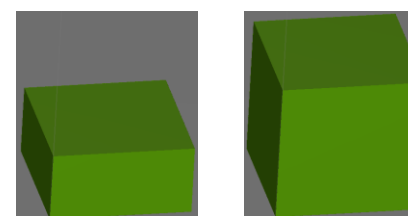
Direction



Vector : 0,0,1

Vector : 1,1,1

Height

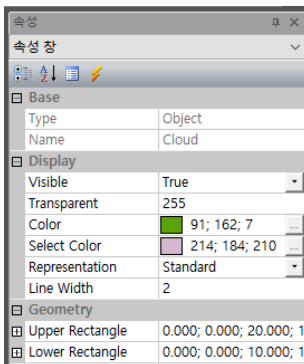


5

10

Rendering : Primitive type : DCube

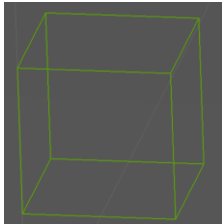
[3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/DCube.h](#)



Attribute

- Visible : 보기/숨김
- Transparent : 투명도
- Color : 색상
- Select Color : 선택 색상
- Representation : 표현 방법
- Line Width : 라인 표현 방법에서 라인의 두께
- Upper Rectangle : 상위 사각형
- Lower Rectangle : 하위 사각형
- Direction : 원의 방향
- Origin : 원의 중점

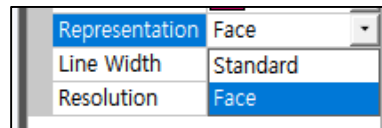
Representation



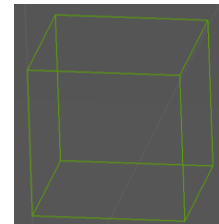
Standard



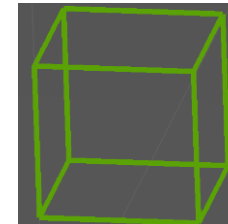
Face



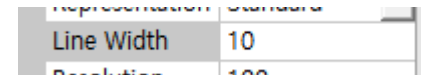
Line Width



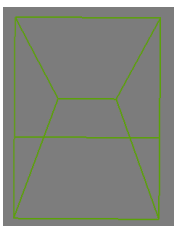
2



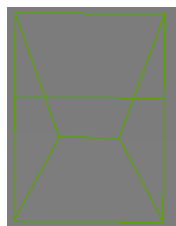
10



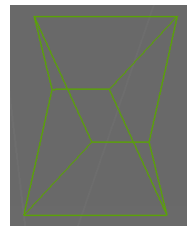
Resolution



상판 사다리꼴



하판 사다리꼴

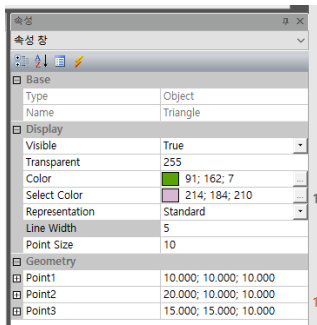


상하판 사다리꼴

Geometry	
Upper Rectangle	0.000; 0.000; 20.000; 10.000;
Point1	0.000; 0.000; 20.000
Point2	10.000; 0.000; 20.000
Point3	10.000; 10.000; 20.000
Point4	0.000; 10.000; 20.000
Lower Rectangle	0.000; 0.000; 10.000; 10.000;
Point1	0.000; 0.000; 10.000
Point2	10.000; 0.000; 10.000
Point3	7.000; 10.000; 10.000
Point4	3.000; 10.000; 10.000

Rendering : Primitive type : DTriangle

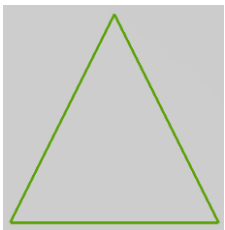
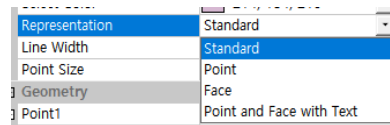
[3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/DTriangle.h](#)



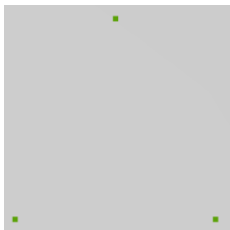
Attribute

- Visible : 보기/숨김
- Transparent : 투명도
- Color : 색상
- Select Color : 선택 색상
- Representation : 표현 방법
- Line Width : 라인 표현 방법에서 라인의 두께
- Point Size : 점 표현 방법에서 점의 크기
- Point1~3 : 삼각형의 각 꼭지점

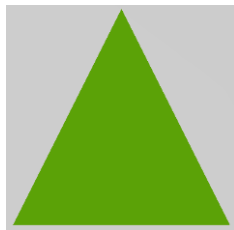
Representation



Standard



Point



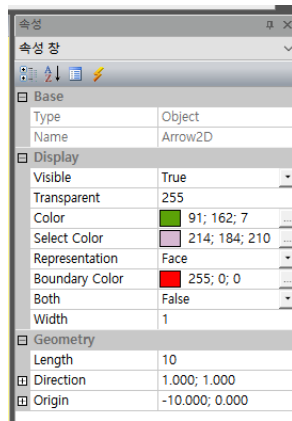
Face

Rendering : Primitive type : DArrow2D

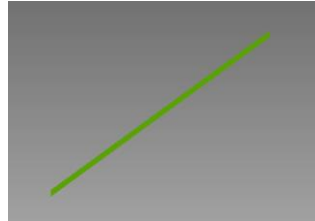
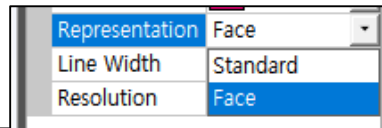
<3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry/DArrow2D.h>

Attribute

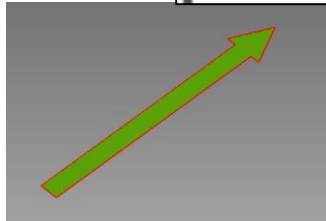
- Visible : 보기/숨김
- Transparent : 투명도
- Color : 색상
- Select Color : 선택 색상
- Boundary Color : 외곽선 색상
- Representation : 표현 방법
- Both : 화살표 위치(FACE 표기 방법 만)
- Width : 화살표의 두께
- Length : 화살표의 길이



Representation

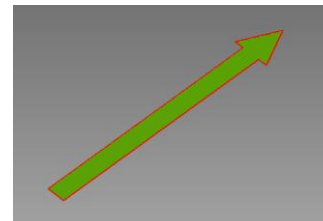
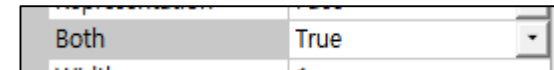


Standard

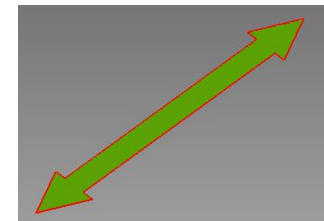


Face

Both

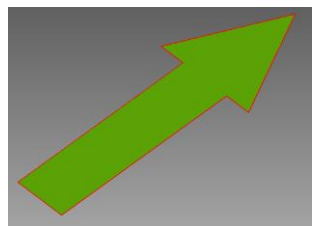
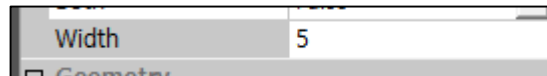


FALSE

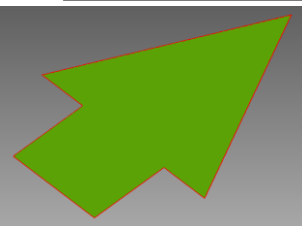


TRUE

Width

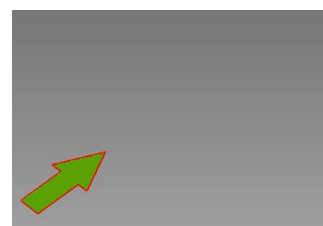
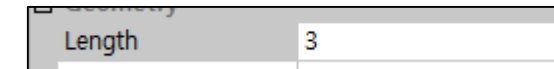


5

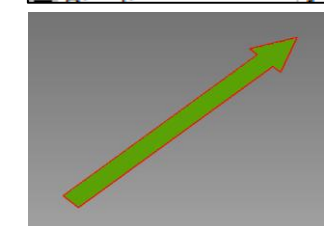


10

Length



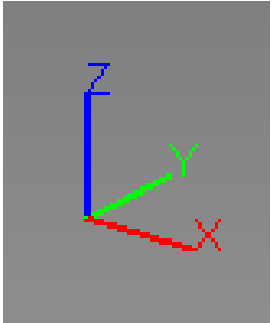
3



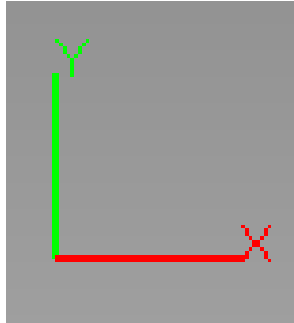
10

Rendering : Primitive type : ETC

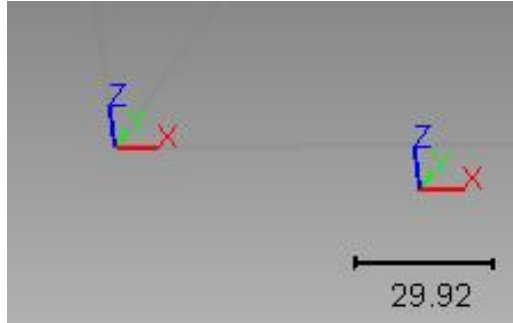
[3DReshaperSDK/Reshaper3DRenderingEngine/Display/Object/Geometry](#)



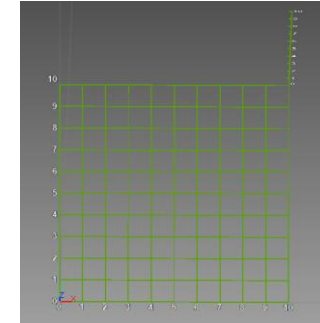
CoordinateSystem



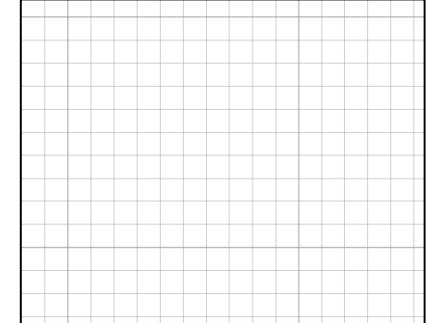
CoordinateSystem2D



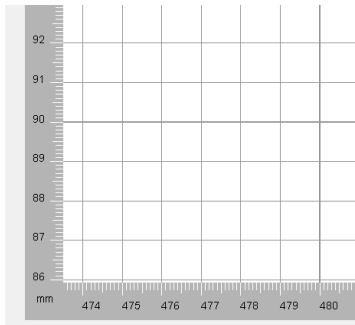
SceneCoordinateSystem



GridRuller



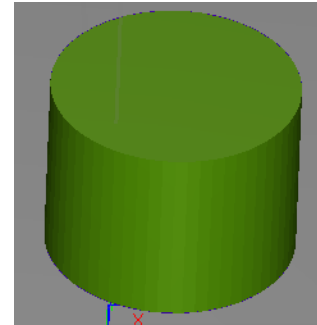
GridDeepenExtend



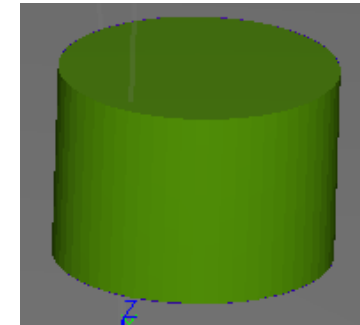
RulerPlatGradation



Ruler

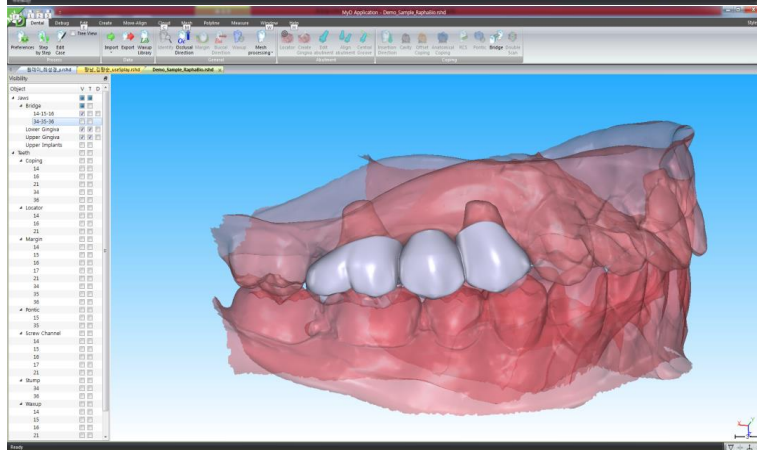
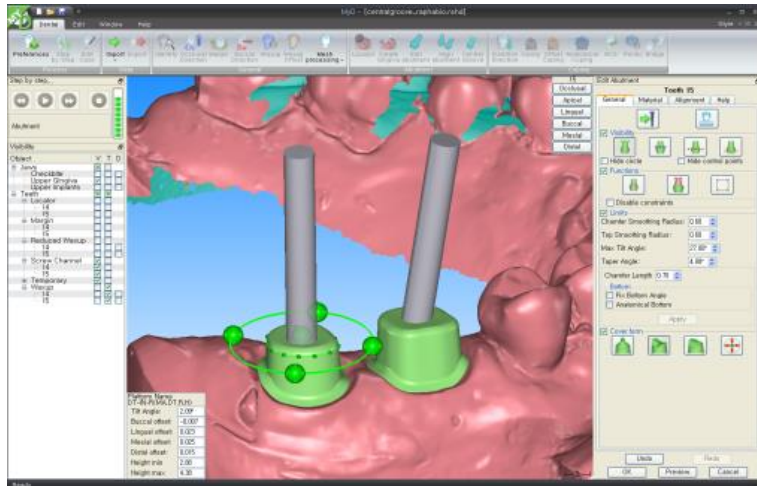


Cylinder

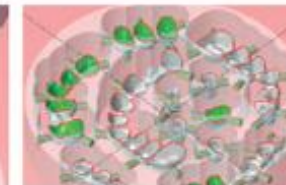
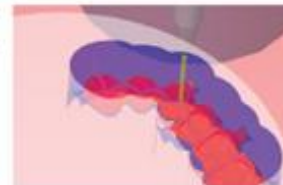
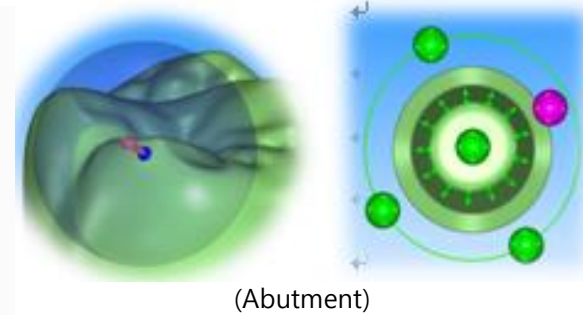
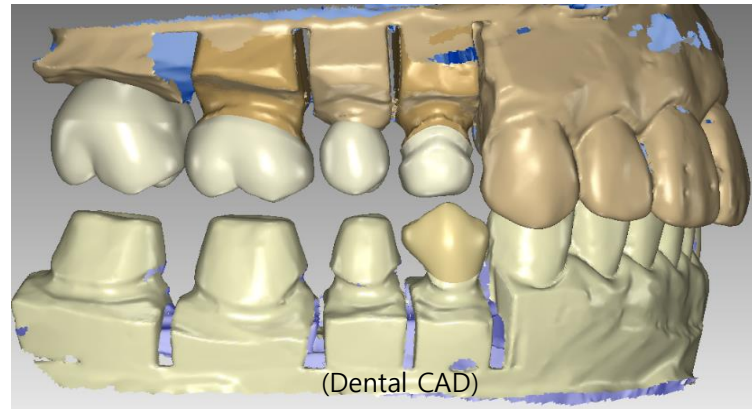


Pipe

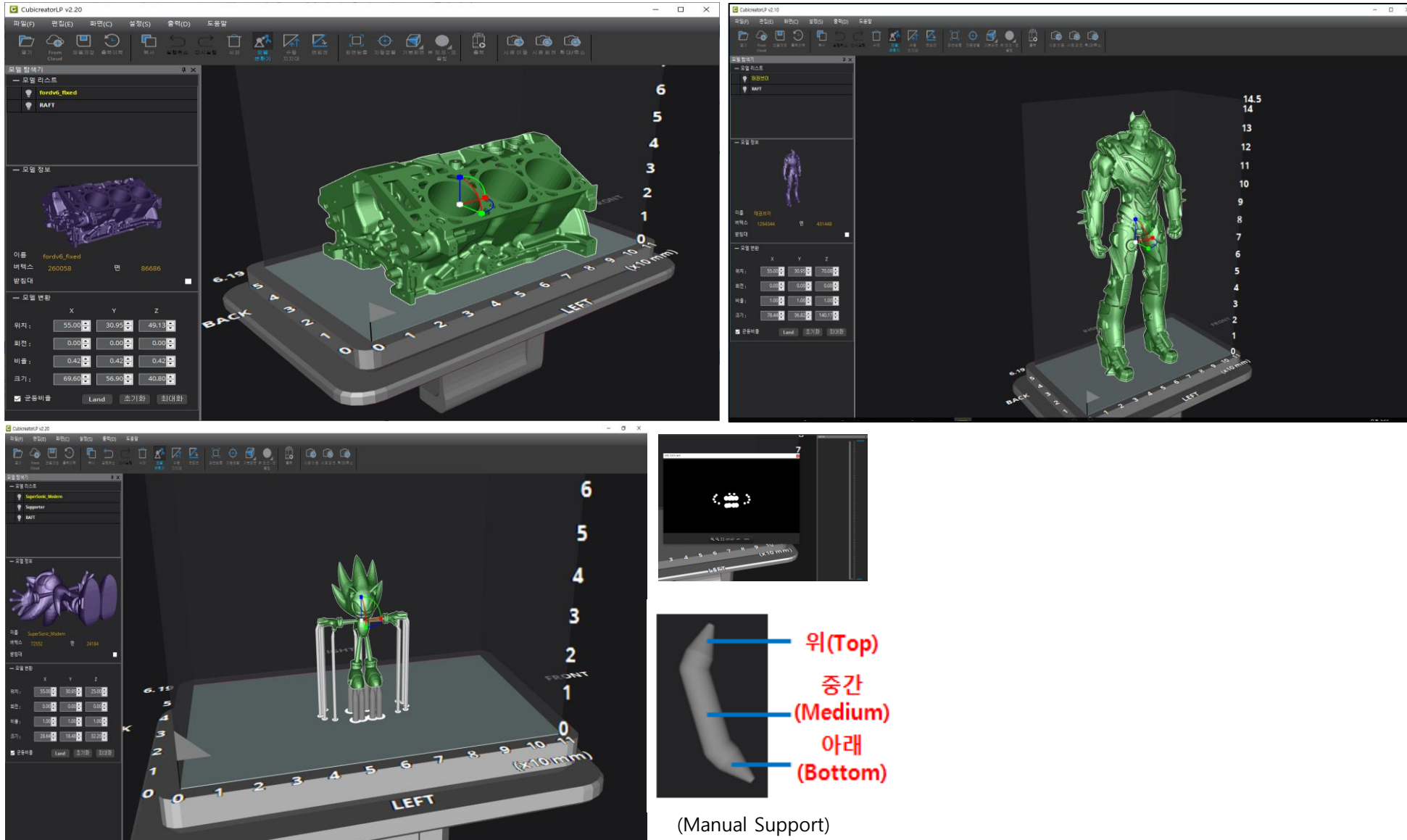
Application : MyD



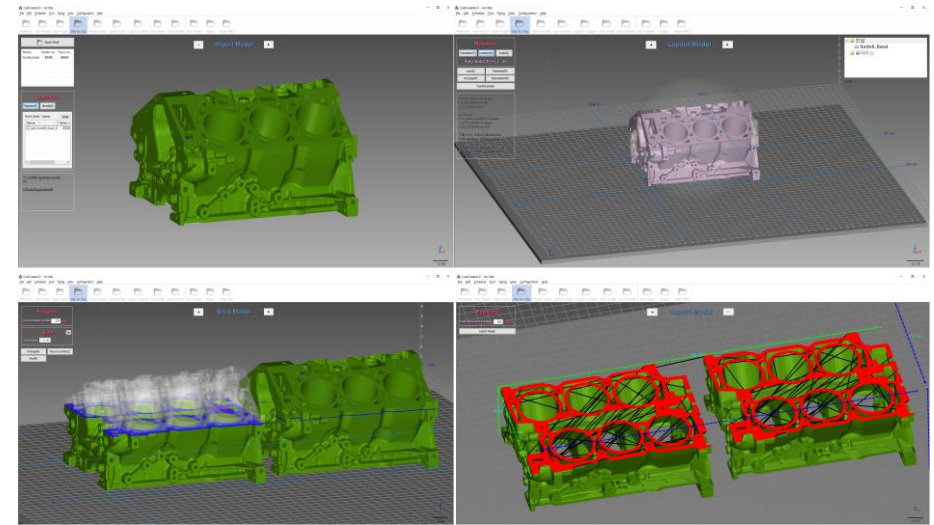
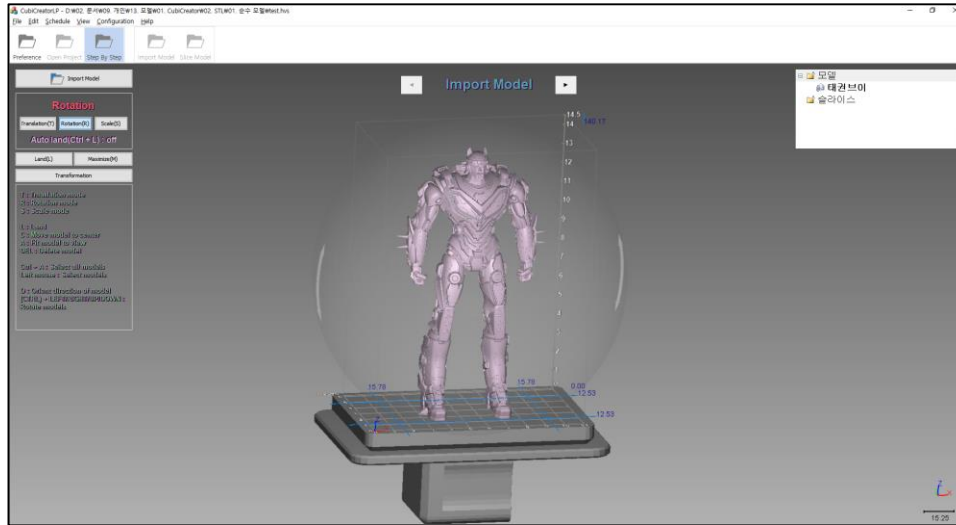
MyPlant



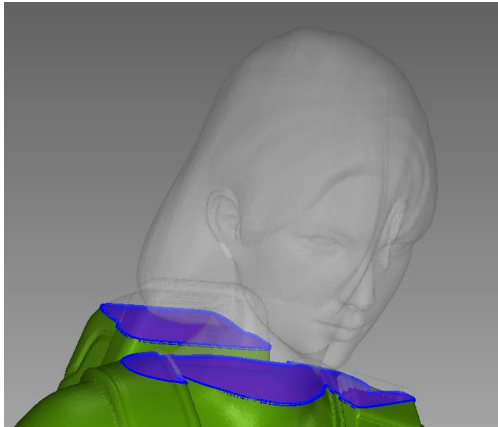
Application : CubiCreatorLP v1



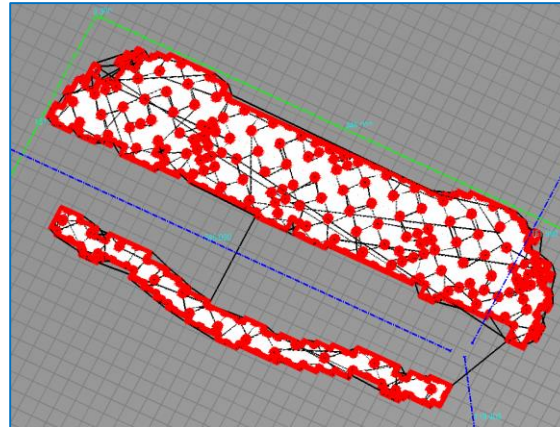
Application : CubiCreatorLP v2



(3D 프린터 준비 소프트웨어)

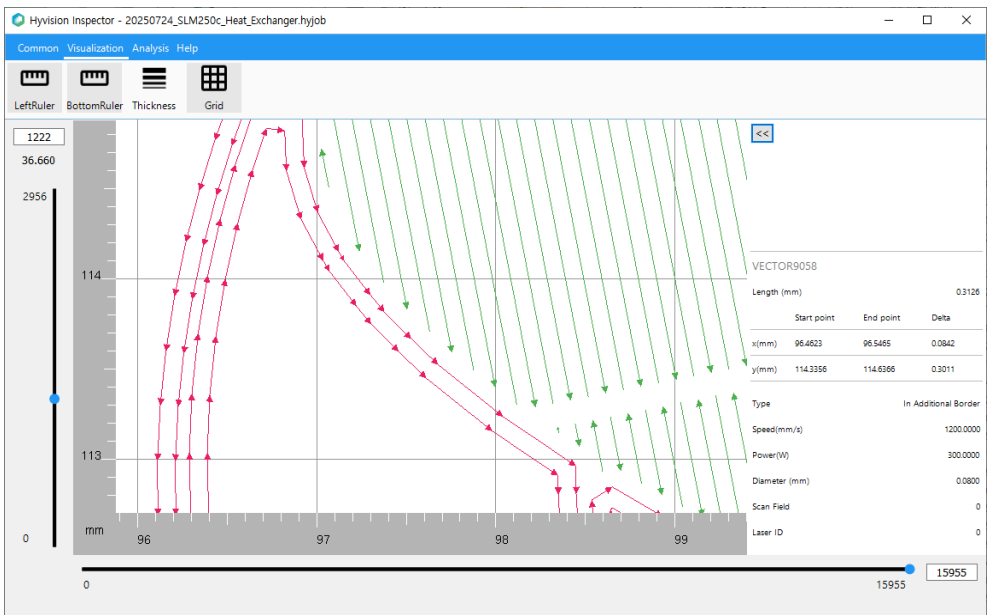


Clipping, Front/Back, Transparency

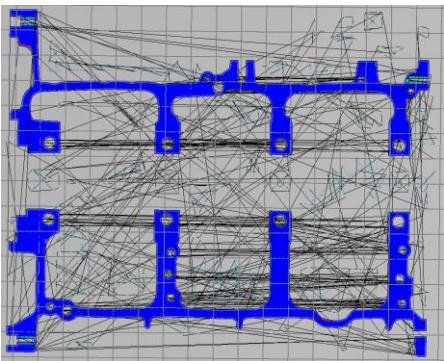
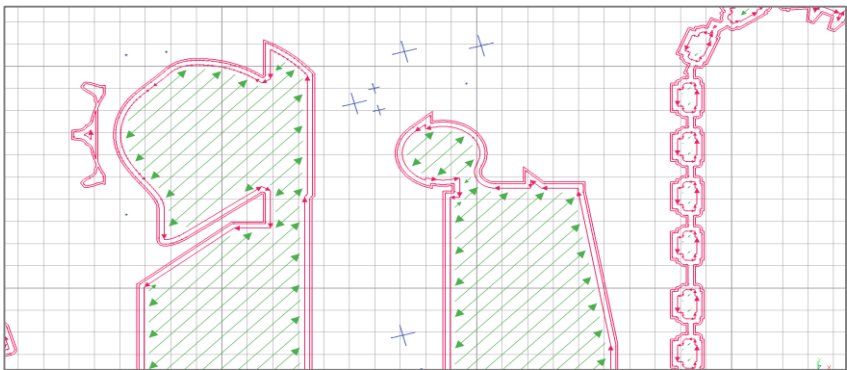


(BuildProcessor For SLA + DLP)

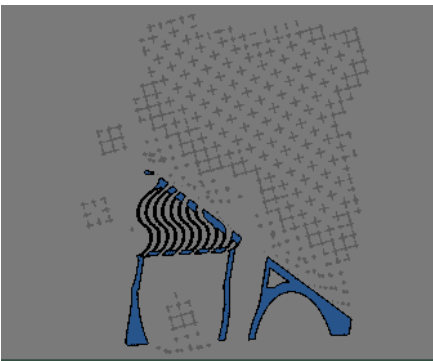
Application : HyInspector



(HyInspector)

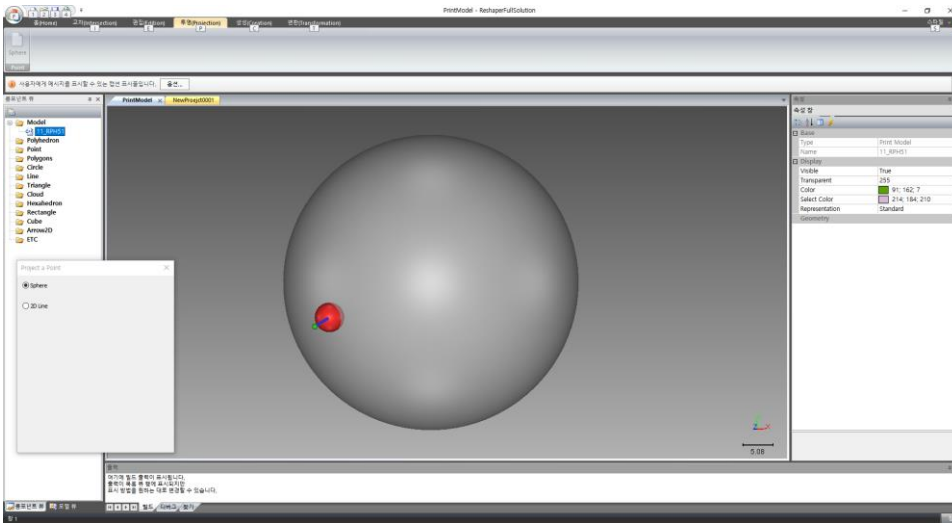


(Inspector)



(Path Viewer)

Application : ReshaperSolution(미완)



Thank you